Project Management for Multi-Component Systems

Coordinating work across HW, SW, and people

SEAtS | 834694

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Intended Learning Outcomes

By the end of this lecture, you will be able to:

1. Apply MoSCoW prioritisation to manage scope and stakeholder expectations.
2. Use T-shirt sizing for early-stage effort and complexity estimation.
3. Compare agile, stage-gate, and hybrid delivery approaches for multi-domain projects.
4. Create Gantt and burn-down charts to monitor progress.
5. Integrate systemic design principles to coordinate scheduling, workload, and feedback loops across subsystems.

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Agenda

- Why Systemic Project Management
- Systems Thinking to Planning
- MoSCoW Prioritisation
- T-shirt Effort Sizing
- Agile in Systemic Projects
- Scheduling with Gantt Charts
- Tracking with Burn-down Charts
- Preventing Burnout
- Integrated Feedback Ecosystem
- Summary

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Why Systemic Project Management

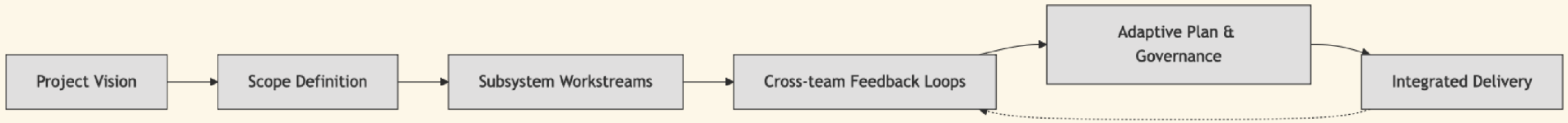
Project management in engineering is more than tasks and timelines—it orchestrates interacting subsystems and the humans who enable them. Traditional linear planning fails when hardware, software, and testing workstreams depend on each other.

⚠ Systemic viewpoint

- Projects behave like adaptive systems with feedback loops and emergent risks.
- Systemic design combines systems thinking with design thinking to handle this complexity.

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Systemic project planning cycles plans through continuous feedback instead of fixed schedules.



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Systems Thinking to Planning

Systemic design integrates high-level vision with detailed interactions.

ⓘ Informed decomposition

- Break large optimisation problems so components improve independently yet converge on system goals.
- Mirrors project work breakdown structures with feedback milestones.

✓ Feedback-aware planning

- View subsystems as nodes in a network.
- Adjust effort, time, and resources using feedback reviews rather than rigid baselines.

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MoSCoW Prioritisation

Not every requirement is equally important; MoSCoW keeps ambitions aligned with limited time and resources.

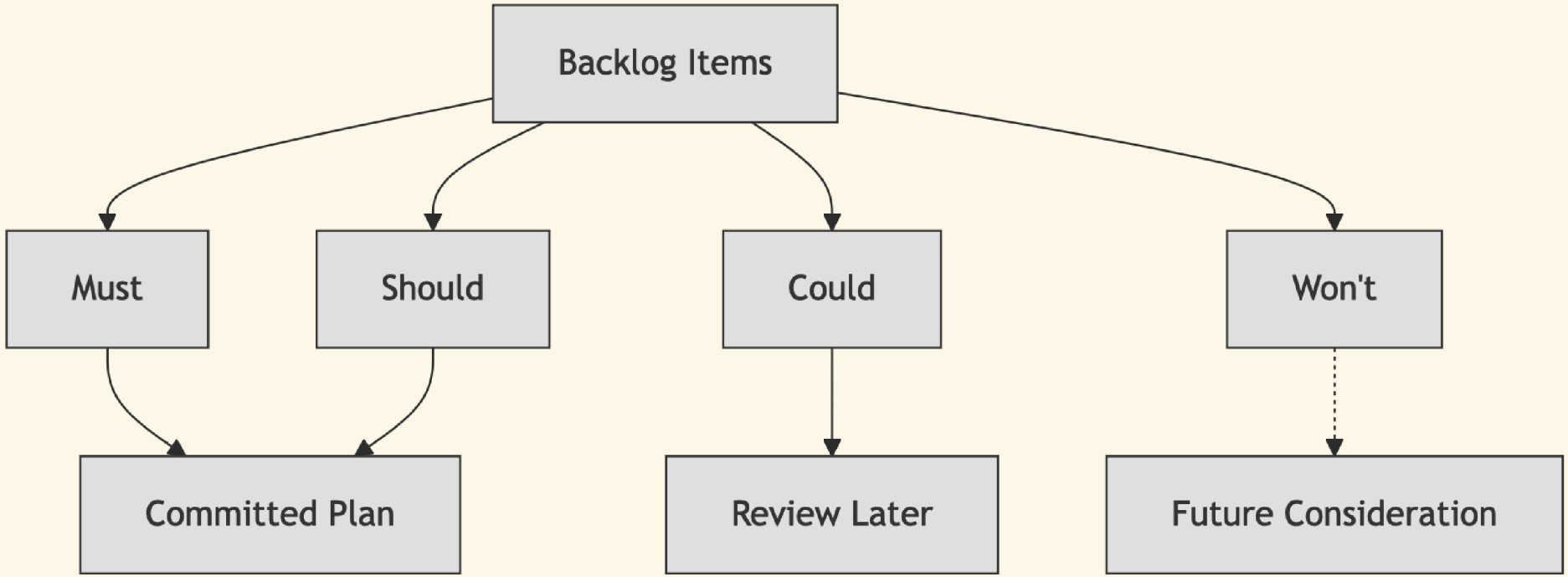
Category	Description	Example (Autonomous Drone)
Must-Have	Essential for success	Stable flight control loop
Should-Have	Important but not critical	Auto-return on signal loss
Could-Have	Nice if time permits	Custom mission planner UI
Won't-Have	Explicitly out of scope	Fully autonomous parcel drop

ⓘ Why it matters

- Prevents scope creep and aligns stakeholders.
- Systemically manages variety: team effort matches system complexity.

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Visualising MoSCoW categories clarifies commitments versus stretch goals.



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T-shirt Effort Sizing

Detailed estimates are unreliable early on; use relative sizing for fast alignment.

Size	Interpretation	Approximate Effort
XS	Trivial task	< 1 day
S	Small module	1–2 days
M	Moderate complexity	3–5 days
L	Complex feature	1–2 weeks
XL	Multi-team, high complexity	> 2 weeks

Subsystem teams (mechanical, software, controls, testing) size their backlogs independently, then harmonise at integration planning.

✓ Systemic alignment

- Relative sizes act like meta-models—fast approximations refined as knowledge grows.
- Supports sequencing and capacity planning before detailed scheduling.

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Agile in Systemic Projects

Definition: Agile is an iterative, incremental way of delivering value where cross-functional teams pull work from a prioritised backlog, ship in short cycles, and adapt plans through frequent feedback.

Agile is one option in a spectrum of delivery models; picking the right blend depends on regulatory load, hardware lead times, and integration cadence.

Approach	Cadence & Structure	Where it shines	Systemic risk when misapplied
Agile / Scrum	1–4 week increments, empirical backlog	Fast-learning domains, software-heavy subsystems	Hardware and compliance teams left out, leading to desynchronised releases
Stage-Gate / Waterfall	Sequential phases with tollgates	Certifiable artefacts, long-lead procurement	Late discovery of integration issues; rework propagates across domains
Hybrid / Wagile	High-level gates with agile delivery inside	Multi-discipline programmes needing both assurance and adaptability	Requires explicit integration reviews or the layers drift apart
Critical Chain / Flow	Buffer-managed, resource-constrained chains	Shared tooling or labs with tight capacity	Buffers erode if feedback signals are ignored

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Choosing the mix

- Map subsystem lead times and regulatory checkpoints; use gates where evidence is mandated, sprints where learning is rapid.
- Synchronise cadences: align sprint reviews with milestone demos so mechanical, firmware, and operations teams stay on the same rhythm.
- Keep artefacts interoperable—MoSCoW priorities, T-shirt sizes, and burn-down data should update whichever method a team follows.

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✓ When agile clicks

- Integrated robotics squad plans two-week “flyable slice” sprints covering firmware, airframe tweaks, and pilot rehearsal.
- Shared Definition of Done requires telemetry review, updated risk register, and demo to safety engineers.
- Sprint reviews feed MoSCoW priorities, so newly surfaced hazards replace lower-value backlog items immediately.

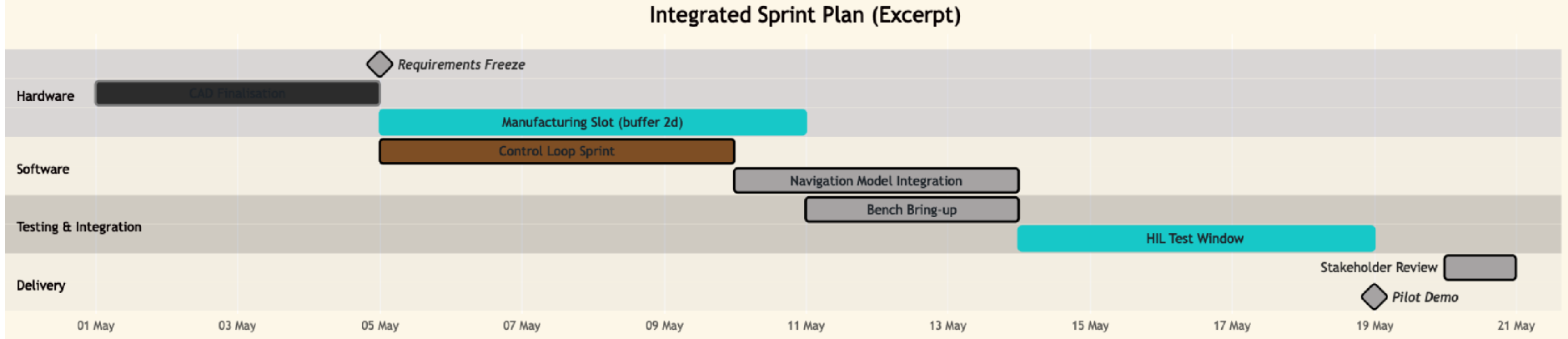
⚠ When agile fails

- Only the software team sprints; hardware, compliance, and training follow quarterly gates, so integrations collapse.
- Stand-ups stay task-focused and siloed, hiding cross-team blockers until just before test windows.
- Velocity targets pressure teams to inflate story points while verification debt and safety cases slip.

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Gantt Charts

Gantt charts map duration, dependencies, and milestones across teams.



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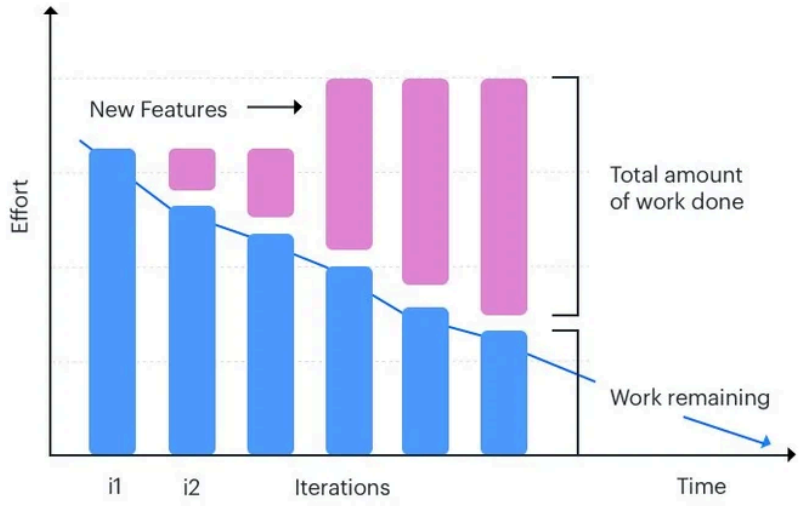
⚠ Coordination cues

- Maintain subsystem-specific Gantts plus a system-level integration view.
- Use milestones and `crit` flags to surface system constraints; keep buffers visible.
- Update schedules through feedback cycles, revising dependencies as learnings emerge.

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Tracking with Burn-down Charts

A burn-down chart shows remaining work versus time—driving continuous feedback between planning and execution.



- Signals resource overload early.
- Shows whether effort distribution aligns with MoSCoW priorities.
- Supports reflection on convergence akin to distributed optimisation monitoring.

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Preventing Burnout

Burn-down charts also surface human stress. Steep slopes before deadlines or early plateaus indicate overload or poor estimates.

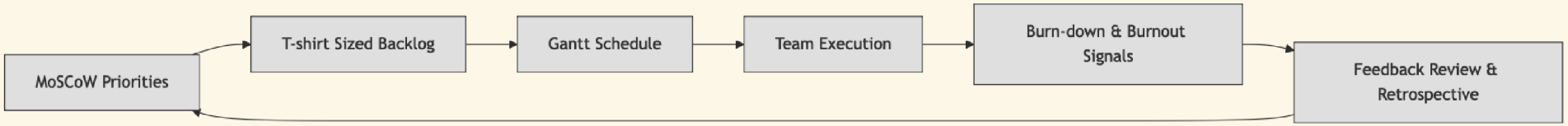
⚠ Human feedback loops

- Run short burnout check-ins during weekly stand-ups.
- Ask “How’s workload and focus?” to catch unsustainable patterns.
- Adjust priorities or reassign tasks based on MoSCoW and T-shirt insights.

Systemic project plans must balance performance with well-being to stay sustainable.

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Project control becomes a feedback ecosystem linking prioritisation, estimation, scheduling, and human signals.



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Integrated Feedback Ecosystem

Systemic project management brings together:

- MoSCoW prioritisation for scope clarity.
- T-shirt sizing for fast estimation.
- Gantt charts for dependency-aware scheduling.
- Burn-down charts and burnout chats for monitoring.
- Feedback reviews to recalibrate continuously.

Together these create a human-technical control loop that adapts to uncertainty across subsystems.

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Summary

- Complex engineering projects require systemic planning, not linear task lists.
- Prioritisation (MoSCoW) and estimation (T-shirt sizing) align stakeholders and resources.
- Gantt and burn-down charts provide visibility into schedule health and workload.
- Delivery approaches must be chosen intentionally—blend agile, gated, and flow methods to match subsystem realities.
- Agile ceremonies add value when they bridge hardware, software, and people workflows.
- Continuous feedback loops sustain both technical progress and team well-being.

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